SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I S. Sai Kumar (192324254) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Sai Kumar

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

ASSESSMENT TOOL-1

Short Answer Test

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Course Code: CSA1514

Course Name: Cloud Computing and Big Data Analytics.

Definition and Key Characteristics of Cloud Computing

Definition:

Cloud computing is the on-demand delivery of computing services - including servers, storage, databases, networking and software - over the internet ("the cloud") using a pay-as-you-go pricing model.

Four Key Characteristics vs On-Premise Computing:

* **On-Demand Self-Service:** Users can provision computing resources (like CPU time and network storage) automatically without requiring human intervention from service providers, unlike on-premise setups that require manual hardware procurement and configuration.
* **Broad Network Access:** Capabilities are available over the network and accessed through standard mechanisms via heterogeneous client platforms (laptops, mobile phones), whereas on-premise access is often restricted to local corporate intranets or dedicated VPN
* **Resource Pooling:** The provider's computing resources are pooled to serve multiple consumers, dynamically assigning physical and virtual resources according to demand, compared to dedicated, single-tenant physical servers in traditional environments
* **Rapid Elasticity:** Automatically scales up or down based on real-time workload demands, preventing idle over-provisioning.

2- Evolution of Cloud Computing:

* Hardware Evolution: Shifted from high-cost mainframes to low-cost, high-density commodity hardware, multi-core processors and blade servers.

* Virtualization Evolution: Introduced hypervisors to decouple operating systems from physical hardware, enabling multi-tenant utilization and workload migration.

* Internet & Distributed Computing: High-speed broadband, cluster computing and grid architectures made coordinated remote execution viable.

* Internet Software Evolution: Modernized monolithic applications into Service-Oriented Architecture (SOA), web services, RESTful APIs, and microservices.

Enabling Modern Cloud Platforms:

Combining virtualization with high-speed internet and standardized web API's allowed providers to automate infrastructure management and deliver computational power as a self-service utility.

3- Server Virtualization vs Cloud Computing

Role:

Server virtualization uses a hypervisor to partition a single physical server into multiple isolated Virtual Machines (VMs), providing the baseline infrastructure efficiency, multi-tenancy and resource isolation that cloud architectures require.

Virtualization VS Cloud Computing:

Dimension	Virtualization (Enabling Technology)	Cloud Computing (Service Model)
Core Concept	Software abstraction separating physical hardware from the OS.	Comprehensive model delivering pooled IT resources as an on-demand service.
Automation	Requires manual administrative configuration and provisioning.	Self-service drive via automated orchestration, self-service portals, and APIs.
Cost Model	Capital Expense (CapEx) focused on hardware and hypervisor licensing.	Operation Expense (OpEx) based on pay-per-use consumption.

4. Role of Data Centers in Cloud Services

* Physical Infrastructure: Houses the clustered servers, high-throughput networks and storage area networks (SANs) that execute cloud workloads.

* Reliability & Uptime: Provides redundant power supplies, backup generators and industrial HVAC cooling to maintain uninterrupted service and fulfill SLAs.

* Global Reach & Latency: Distributes physical sites across global regions and availability zones to deliver low-latency access and real-time disaster recovery.

* Physical security: Protects multi-tenant hardware through controlled perimeter access, 24/7 surveillance and automated fire suppression systems.

Service Models : IaaS, PaaS, SaaS and XaaS

* IaaS (Infrastructure as a Service):

Provides raw computing, storage and networking, usage users manage the OS, runtime and applications.

Example : Amazon EC2

* PaaS (Platform as a Service):

Provides a pre-configured environment with managed OS, middleware and database engines, users only deploy and manage application code.

Example: Google App Engine

* SaaS (Software as a Service):

Delivers fully managed, ready-to-use software applications accessible via a web browser.

Example: Microsoft 365.

* XaaS (Anything / Everything as a Service):

An umbrella term for any distinct IT function delivered as an on-demand, managed network service.

Example: AWS Lambda (Function as a Service / FaaS)

10/10/25